WASP-43b

R-0373

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_250410` · created 2026-07-18T17:17:29+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460775.772 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.026
Transit depth (Rp/Rs)^2	1.91 +/- 0.72 [%]
Orbital Inclination (inc)	86.17 +/- 1.95
Airmass coefficient 1 (a1)	0.943 +/- 0.018
Airmass coefficient 2 (a2)	0.0301 +/- 0.0086
Scatter in the residuals of the lightcurve fit is	1.93 %
Best Comparison Star	#3 - [404, 370]
Optimal Method	PSF photometry
Transit Duration (day)	0.0577 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.138 ± 0.026 vs 0.1594 (deviation 0.82σ)
Duration fit vs published	0.0577 d vs 0.0512 d (deviation 2.02σ)
Mid-time O–C	-6.4 min
Depth SNR	5.3
Residual scatter	1.93 %

Light curve

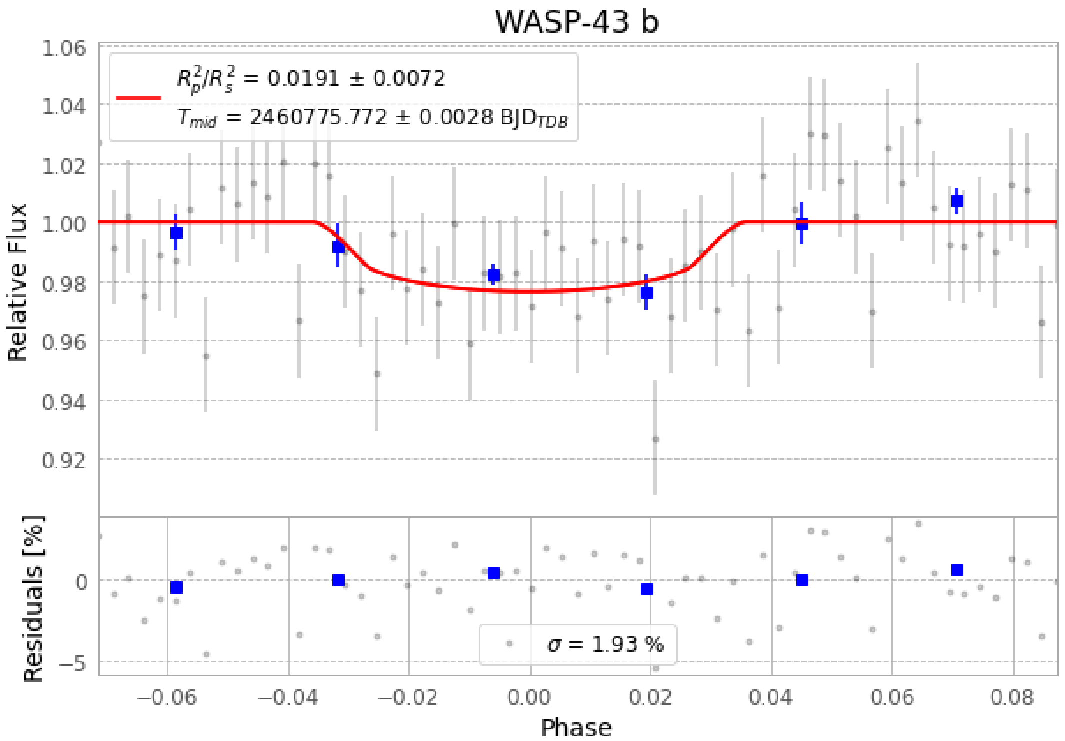

Figure 1 — FinalLightCurve_WASP-43 b_2025-04-09.png (SHA-256 f367b3203b1501c4…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [383, 243], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 06d1a7abcd970dba…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit c09d8bf`

Data provenance

63 FITS frames (41 MB), `WASP-432504100050016.FITS` → `WASP-432504100080616.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250410.log (9a5f3e3aa0f7…), FinalLightCurve_WASP-43 b_2025-04-09.png (f367b3203b15…), FinalLightCurve_WASP-43 b_2025-04-09.csv (977844528bad…)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)